

Academics

Ph.D. Department of Physics, Kurukshetra University, Kurukshetra

2020

Thesis “*Ion Implantation Induced Growth of Silver nanoparticles in Glass*”

Research Interests

Materials having dimensions in the nano regime are attracting considerable attention due to their interesting chemical and physical characteristics owing to increased surface to volume ratio and quantum confinement effect. Among numerous types of nanoscale materials, innovative functionalities associated with metallic nanostructures let them occupy a special place in electrical, optical and medical industry leading to increasing interest of researchers around the world. Nanostructures of noble metals like silver (Ag), copper (Cu) and gold (Au) have a peculiar property of interacting with the incident electromagnetic radiation through the process of surface plasmon resonance (SPR) that results into a strong absorption of the incident radiation in visible region. Unique features of these metallic nanostructures can be exploited to maximum extent by encapsulating them in a suitable transparent dielectric medium which not only reduces their degree of agglomeration but also protect them against contamination from the external environment and thus transparent silicate glasses such as soda lime glass are remarkable contenders for this purpose.

Current work has been performed to investigate the morphological, structural, optical and electrical behavior of soda lime glass embedded with Ag nanoparticles synthesized by employing ion-exchange ($\text{Na}^+ \leftrightarrow \text{Ag}^+$) treatment followed by Argon implantation at variable angles of incidence (normal, 30°, 40° and 50°) using transmission electron microscopy (TEM), field emission scanning electron microscopy (FESEM), glancing incidence X-ray diffraction (GIXRD), atomic force microscopy (AFM), fourier transform infra-red spectroscopy (FTIR), Raman spectroscopy, Rutherford back scattering spectrometry (RBS), X-ray photoelectron spectroscopy (XPS), UV-Visible spectroscopy, photoluminescence spectroscopy and current-voltage (I-V) measurements.

Publications

1. **Sonal**, Annu Sharma, “*Energetic argon beam stimulated growth of plasmonic silver nanoparticles in Ag^+ -exchanged soda glass: A study on the structural, optical, photoluminescence and electrical behavior*”, Materials Science and Engineering B 263 (2021) 114860.

2. **Sonal**, Annu Sharma, “Argon beam prompted alterations in morphological, structural, optical and electrical properties of $\text{Na}^+ \leftrightarrow \text{Ag}^+$ ion-exchanged glass”, Materials Chemistry and Physics 257 (2021) 123795.
3. **Sonal**, Annu Sharma, Sanjeev Aggarwal, “200 keV Ar^+ induced growth of silver nanoparticles in soda lime glass”, Journal of Integrated Ferroelectrics 205 (2020) 156.
4. **Sonal**, Annu Sharma, Sanjeev Aggarwal, “Silver glass nanocomposites synthesised by 200 keV Ar^+ ions at oblique incidence for potential applications as filters in mid infrared region”, Optical Materials 84 (2018) 807.
5. **Sonal**, Annu Sharma, Sanjeev Aggarwal, “Optical investigation of soda lime glass with buried silver nanoparticles synthesized by ion implantation”, Journal of Non-Crystalline Solids 485 (2018) 57.
6. **Sonal**, Annu Sharma, Sanjeev Aggarwal, “Optical characterization of Ag-glass nanocomposites using Mie theory”, AIP Conference Proceedings 2093 (2019) 020004.
7. **Sonal**, Annu Sharma, Sanjeev Aggarwal, “Effect of embedded silver nanoparticles on refractive index of soda lime glass”, AIP Conference Proceedings 1953 (2018) 030114.

Research Papers Presented in International/National Conferences

1. 2nd International Conference on Recent Advances in Applied Physics and Material Science (RAM 2024) held at Bikaner Technical University, Bikaner, 15-16 November 2024, “Silver Glass Nanocomposites based UV Bandpass Filters”, **Sonal**, Annu Sharma.
2. International Conference on Nano-Structured Materials and Devices (ICNSMD-2018) held at University of Delhi, Delhi, 17-20 December 2018, “200 keV Ar^+ induced growth of silver nanoparticles in soda lime glass”, **Sonal**, Annu Sharma, Sanjeev Aggarwal.
3. National Conference on Recent Advances in Condensed Matter Physics (RACMP 2018) held at Kurukshetra University, Kurukshetra, Haryana, 12-13 October 2018, “Optical Characterization of Ag-Glass Nanocomposites Using Mie Theory”, **Sonal**, Annu Sharma, Sanjeev Aggarwal.
4. 5th International Conference on Ion Beams in Materials Engineering and Characterizations (IBMEC 2018) held at Inter-University Accelerator Centre, New Delhi, 9-12 October 2018, “Structural and Optical analysis of Ag-Glass Nanocomposites for Potential Optical Applications”, **Sonal**, Annu Sharma, Sanjeev Aggarwal.
5. 2nd International Conference on Condensed Matter and Applied Physics (ICC-2017) held at Govt. Engineering College, Bikaner, Rajasthan, 24- 25 November 2017, “Effect of

Embedded Silver Nanoparticles on Refractive Index of Soda Lime Glass", **Sonal**, Annu Sharma, Sanjeev Aggarwal.

6. 4th International Conference on Nano Structuring by Ion Beams (ICNIB-2017) held at Devi Ahilya University, Indore, Madhya Pradesh, 11-13 October 2017, "*Engineering Optical Parameters of Glass by Embedding Silver Nanoparticles Using Low Energy Ions at Oblique Incidence*", **Sonal**, Annu Sharma, Sanjeev Aggarwal.
7. International conference on Ion Beams in Materials Engineering and Characterization (IBMEC-2016) held at Inter-University Accelerator Centre, New Delhi, 28 September-1 October 2016, "*Study of optical properties of Ag – glass nanocomposites synthesised by normal and oblique incidence*", **Sonal**, Annu Sharma, Sanjeev Aggarwal.
8. National Conference on Current Developments in Physics (CDIP-2016) held at S. D. (PG) College, Panipat, Haryana, 28-29 March 2016, "*UV-visible characterization of Ag-glass nanocomposites synthesized by Ar⁺ implantation*", **Sonal**, Annu Sharma, Sanjeev Aggarwal.
9. National Conference on Photonics and Material Science (NCPMS-2015) held at Guru Jambheshwar University, Hisar, Haryana, 18-19 November 2015, "*Study of optical properties of Ag-glass nanocomposites synthesised by Ar⁺ implantation*", **Sonal**, Annu Sharma, Sanjeev Aggarwal.
10. National Conference on Physics Industry Interface (NCPII-2015) held at Kurukshetra University, Kurukshetra, Haryana, 2-4 September 2015, "*Study of Dielectric Properties of Starch-Ag Nanocomposite*", Meena, **Sonal**, Annu Sharma, Sanjeev Aggarwal.

International / National Schools/ Workshops and Courses Attended

1. Science Academics' Refresher Course in Experimental Physics held at Department of Physics, Kurukshetra University, Kurukshetra, 13-28 September 2015.
2. Two days National Workshop on Ion Beam Induced Growth and Engineering of Materials, held at Department of Physics, Kurukshetra University, Kurukshetra, 4-5 March 2016.
3. International School on Ion Beams in Materials Science, held at Inter-University Accelerator Centre, New Delhi, 22-27 September 2016.
4. One day Workshop on Intellectual Property Rights organized by IPR Cell, Kurukshetra University, Kurukshetra, 21 December 2016.
5. Two days Workshop on Opportunities and Challenges for Women in STEM organized by Women Faculty of Science, Life Sciences, Engineering and Technology, Kurukshetra University, Kurukshetra, 24-25 January 2018.

Teaching Experience

August 2024-Present	Assistant Professor, Maharaja Surajmal Institute of Technology, New Delhi.
December 2022-July 2024	Assistant Professor, Guru Tegh Bahadur Institute of Technology, New Delhi.

Other Examinations Qualified

December 2014 CSIR-UGC NET

2014, 2015, 2017 GATE

Training and FDPs

1. FDP on Developing Healthy Work Culture at Workplace conducted by NITTTR Chandigarh at Maharaja Surajmal Institute of Technology, New Delhi, 1-5 December 2025.
2. FDP on Effective Teaching - Learning Using Social Media conducted by NITTTR Chandigarh at Maharaja Surajmal Institute of Technology, New Delhi, 26-30 May 2025.
3. FDP on Online Learning and Evaluation conducted by NITTTR Chandigarh at Maharaja Surajmal Institute of Technology, New Delhi, 2-6 December 2024.
4. FDP on Outcome-based Education System and Accreditation organized by Guru Tegh Bahadur Institute of Technology, New Delhi, 8-12 July 2024.